Problem Chosen	2023 MCM/ICM Summary Sheet	Team Control Number
C		2621535

Balancing Act: A Quantitative Study of Judge-Audience Dynamics and a Novel Scoring System for Dancing with the Stars

Summary

This paper conducts a comprehensive mathematical modeling study to analyze and optimize the scoring and elimination mechanisms of Dancing with the Stars.

First, to estimate unpublicized audience vote counts, we developed two optimization models. For seasons using a percentage-based synthesis rule (Seasons 3-27), a Linear Programming (LP) model was formulated. For seasons using a ranking-based rule (Seasons 1-2, 28-34), a Mixed-Integer Linear Programming (MILP) model was constructed. Both models infer weekly audience vote shares from historical judge scores and elimination outcomes, demonstrating over 99% accuracy in reproducing past eliminations upon validation. Uncertainty was quantified via sensitivity analysis and Bootstrap methods.

Second, using the estimated votes, we performed a comparative analysis of the two elimination mechanisms. Statistical tests, including Cohen's Kappa, revealed a 78.2% agreement rate, indicating moderate consistency with significant divergence. Spearman correlation analyses proved that the percentage system grants audience votes stronger independent influence. We introduced a "Protection Index," showing the ranking system better protects technically proficient contestants who receive low initial judge scores. Simulations of the "Judges' Save" hybrid mechanism demonstrated its role in correcting potential biases.

Third, we investigated how participant characteristics influence outcomes through multiple linear regression. Models predicting final placement, judge scores, and audience vote share revealed key factors: age has a universal negative effect; a dance partner's professional experience boosts judge scores but reduces audience support, highlighting a preference divergence; and a modeling background leads to consistently lower evaluations from both groups.

Finally, synthesizing these insights, we designed a novel Dynamic Bayesian Balanced System (DBBS). It dynamically calculates a "Technical Value" and a "Popularity Value" for each contestant, updated weekly via Bayesian recursion. A personalized weight, adjusted by competition stage, performance trend, and the value gap, synthesizes them into a final score, with an enhanced "Judges' Save" as a safety net. Historical backtesting across all 34 seasons confirms DBBS outperforms existing systems, achieving the highest prediction accuracy (86.7%), the lowest controversy index, and the best composite score, successfully balancing technical fairness with audience engagement.

Keywords: Voting Estimation; Linear Programming; Mechanism Comparison; Regression Analysis; Dynamic Bayesian System

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1 Introduction

1.1 Problem Background

"Dancing with the Stars", as a globally popular dance competition reality show, has attracted the attention of countless viewers since its debut. In the program, celebrities pair up with professional dancers and are judged by a panel of judges and voted on by the audience. The combination of the two determines the fate of the contestants. The judges' scores are usually aimed at objectively evaluating the technical skills and expressiveness of the dance, while the audience votes are often more subjective, influenced by the personal charm of the stars, their public influence, and even off-stage topics. This mechanism of combining scores and votes not only adds dramatic tension to the show but also sparks discussions about how to balance professionalism and public preference. Historically, the show's producers have adjusted the way judges' scores and audience votes are combined multiple times - initially using a ranking system, then switching to a percentage system, and later reverting to the ranking system due to controversy. Even in specific seasons, a segment was introduced where judges decided the elimination of the bottom-ranked contestants. These changes reflect both the pursuit of fairness and the delicate balance the producers must strike between maintaining the show's entertainment value and ensuring competitive fairness. With the accumulation of data from dozens of seasons, we now have the opportunity to re-examine this mechanism through mathematical models, explore the underlying patterns in the voting, and consider how to design an evaluation system that is both fair and enhances the show's appeal.

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Task 1:** Develop an audience voting estimation model to predict the number of audience votes for contestants and provide a reliability measure for the estimation results.
- **Task 2:**Compare the two score synthesis methods based on the estimated audience voting data, explore the differences in their results, and analyze which method is more inclined to amplify the impact of audience voting.
- **Task 3:**Analyze controversial cases, evaluate the adjustment of competition formats, and assess the impact of introducing new mechanisms on competition results.
- **Task 4:** Put forward mechanism suggestions for future seasons based on the analysis results.
- **Task 5:**Develop a model to analyze the impact of factors such as athlete

characteristics on competitive performance and outcomes.

- **Task 6:**Design a more equitable score synthesis system and elaborate on its value.

1.3 Our Work

To systematically address the aforementioned issues, this paper constructs a complete analytical framework: First, the audience voting distribution is estimated through data preprocessing, Linear Programming (LP) and Mixed Integer Linear Programming (MILP) models, with reliability verification conducted accordingly. Second, based on the estimation results, the differences between the two elimination mechanisms are compared via statistical tests and quantitative indicators, and the impacts of controversial cases and the "Judges Save" mechanism are analyzed. Third, a multiple regression model is used to analyze the differentiated impacts of contestant characteristics on scores and votes. Finally, integrating Bayesian updating and dynamic weight adjustment, a novel score fusion system (DBBS) is proposed, and its fairness, stability and practicality are verified using historical data.

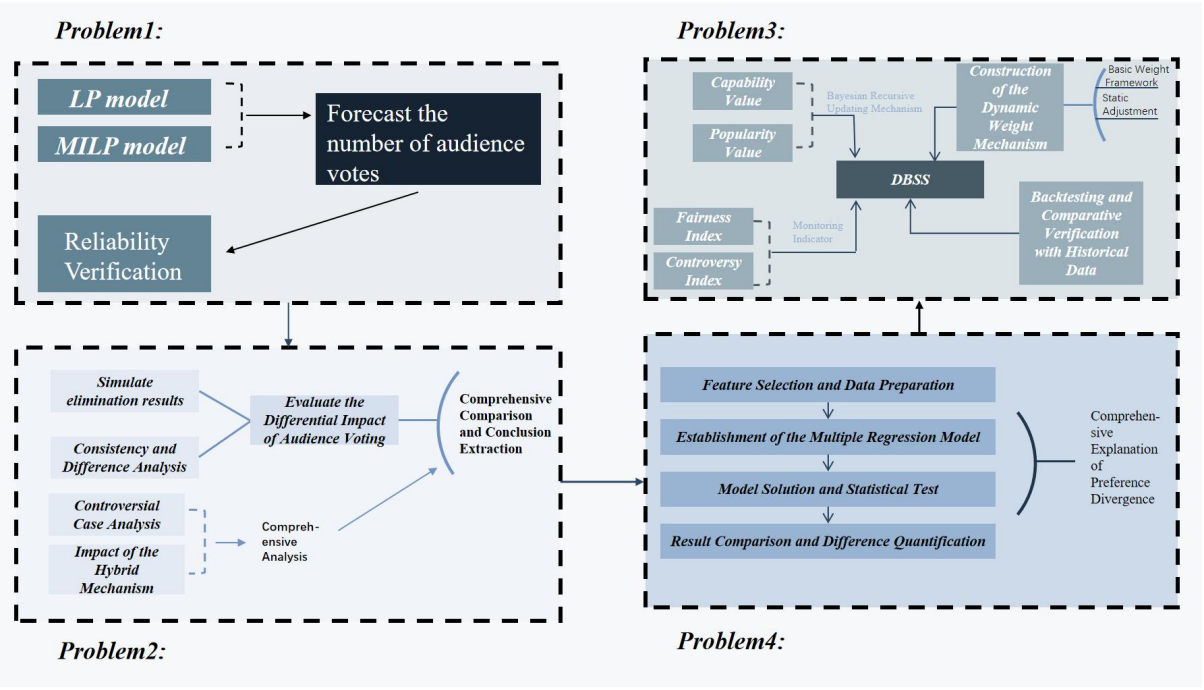


Figure1：The Flow Chart of Our Work

2 Assumptions and Justifications

Some fundamental assumptions are listed below.

- **Assumption1：** We assume that the total number of audience votes per week is a constant.Then calculate the proportion of fans' votes in the total votes.

→Justification： Under the percentage system, we need to calculate the proportion of audience votes rather than their absolute numbers. In this case, as long as we assume there is a common constant denominator for the total votes (regardless of the specific value), the vote proportion for each contestant can be derived through normalization.Even if the actual total number of votes changes week by week, the elimination results will remain

unchanged as long as the vote distribution stays the same.

- **Assumption2：** Fan voting is influenced by subjective attributes such as contestants' popularity, fan base and topicality, and is partially independent of judge scores.

→Justification： Judge scores are primarily based on technical and objective criteria for dance performance, while fan voting is more easily driven by subjective factors such as contestants' pre-existing popularity, audience personal preferences, media exposure, and emotional identification. Although the two are somewhat correlated, there are fundamental differences in their evaluation dimensions and decision-making mechanisms. Therefore, the assumption of their partial independence is not only consistent with real-world observations but also enables the model to more accurately isolate the impacts of different factors.

- **Assumption3：** Perturbation analysis is sufficient to reflect the uncertainty of estimation.

→Justification： It is assumed that the uncertain interval of vote estimation can be effectively quantified through the bootstrap simulation of perturbations in judge scores and fan share.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
i, w, s	Indices for contestant, week, and season	Positive integers
$S_{i,w}$, $\hat{f}_{i,w}$	Judge's total score and estimated audience vote share for contestant i in week w	$S_{i,w} \geq 0$, $\hat{f}_{i,w} \in [0,1]$
$P_{i,w}$	Composite score under the percentage system	[0,2]
$C_{i,w}$	Composite rank value under the ranking system (higher = worse)	Positive integer
κ	Cohen's Kappa coefficient (agreement measure)	$[-1,1]$
D_w, δ_s	Divergence index and season disagreement rate	[0,1]
$\pi_{i,w}$	Protection index for contestant i in week w	$\mathbb{R}$
B_i, J_i	Events: contestant in bottom two, and judges choose to eliminate them	Boolean
Y_{place}, Y_{judge}, Y_{fan}	Outcome variables: final placement, judge score, fan share	Various
X_{age}, X_{exp}, X_{win}	Predictors: contestant age, partner experience, partner wins	Years/Count
T_{ij}, P_{ij}	Technical and Popularity values in DBBS	Continuous
w_{ij}	Dynamic judge-score weight in DBBS	[0.3,0.7]
C_{ij}^{DBBS}	Final composite score under DBBS	Continuous
F_i, C_i	Fairness Index and Controversy Index for week i	$\mathbb{R}$, [0,1]
T_{ij}^{raw}, P_{ij}^{raw}	Raw Technical and Popularity values (before recursion)	Continuous

σ_{ij}	Standard deviation of judge scores for contestant j in week i	≥ 0
β, γ, α	DBBS parameters: penalty, technical impact, and smoothing coefficients	Positive real
w_{ij}	Dynamic judge-score weight in DBBS	[0.3,0.7]
R	Technical advantage ratio used in Judges' Save decision	$\mathbb{R}$

4 Audience Voting Estimation

4.1 Data Preprocessing

We conducted systematic preprocessing on the complete dataset of *Dancing with the Stars* Seasons 1 to 34 to lay a foundation for subsequent modeling.

- We filtered valid data by season and week, excluding records of eliminated contestants to ensure all analysis subjects were active participants in the corresponding period.
- For missing values in the data, we adopted differentiated processing based on the structural characteristics of the show: natural missing values arising from varying weekly durations across seasons were excluded, while individual missing judge scores were imputed with the average score of other judges in the same week.
- We calculated the proportion of each contestant's total score from the judges, eliminated differences in scoring scales through normalization, and obtained accurate judge rankings using a dual sorting algorithm.
- Contestants were divided into eliminated and non-eliminated groups according to the weekly elimination results, which served as a key basis for the constraints of subsequent models.

4.2 LP：the Percentage Model

For the seasons adopting the percentage-based synthesis rule (Seasons 3 to 27), we have established a linear programming model to estimate the distribution of audience votes. Under this rule, the weekly elimination results are determined by the sum of the proportion of judges' scores and the proportion of audience votes, with the contestant having the lowest composite score being eliminated[1].We assume that f_i denote the proportion of audience votes for contestant i，which is a continuous decision variable to be estimated, satisfying $f_i \geq 0$ and $\sum_{i=1}^{n} f_i = 1$ .The core constraints of the model derive from the elimination logic: for any eliminated contestant $e \in E$ and any non-eliminated contestant $s \in S$，their composite scores must satisfy the strict inequality:

$$S_e + f_e \leq S_s + f_s - \varepsilon \tag{1}$$

where ε is an infinitesimally small positive number(typically set to 10^{-4}), which is used to ensure the strictness of the inequality and avoid ambiguities arising from tied eliminations. The design of the objective function comprehensively considers the rationality and robustness of the estimation and consists of two components: the first component measures the proximity between the distribution of audience votes and the distribution of judges' scores, and the second component acts as a regularization term to discourage overly extreme vote distributions. Its specific form is as follows:

$$\min \sum_{i=1}^{n} (f_i - S_i)^2 + \lambda \sum_{i=1}^{n} \left(f_i - \frac{1}{n}\right)^2 \tag{2}$$

In the formula, λ denotes the regularization coefficient, which is set to 0.1 through cross-validation. This problem constitutes a convex quadratic programming problem. We construct the model using the CVXPY library and call an efficient convex optimization solver to solve it, thus obtaining the estimated values of the audience vote proportion $\hat{f}_i$ for each contestant.

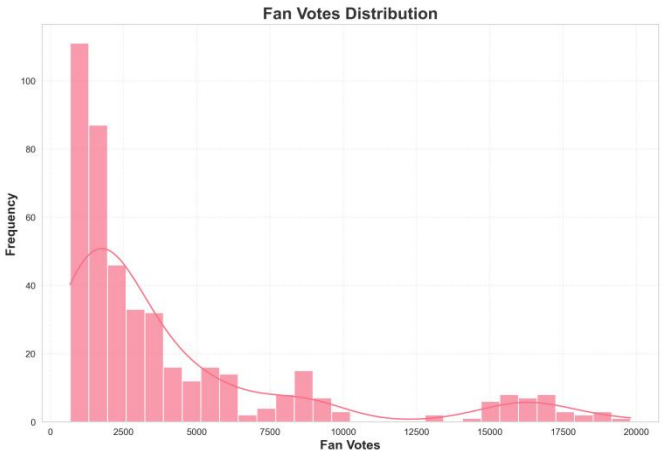
4.3 MILP：the Ranking-based Model

For the seasons adopting the ranking-based synthesis rule (Seasons 1, 2 and all seasons after Season 28),we constructed a mixed-integer programming model based on the Big M method[2]. Under this rule,he weekly elimination is determined by the sum of the judges' ranking R_i^s and the audience vote ranking R_i^f,, with the contestant having the highest total ranking value (the worst performance) being eliminated.The complexity of the model lies in that the audience vote ranking R_i^f is an implicit function determined by the to-be-estimated vote proportion f_i .To handle this nonlinear relationship, we introduce a binary variable $y_{ij} \in \{0,1\}$ to indicate whether the vote proportion of contestant i is no less than that of contestant j：

$$y_{ij} = \begin{cases} 1, & if\ f_i \geq f_j \\ 0, & otherwise \end{cases} \tag{3}$$

4.4 Result Presentation and Reliability Analysis

After solving the model, we obtained the estimated audience vote share $\hat{f}_i$ for each contestant in each week of all seasons. To visually present the estimation results, we generated audience vote distribution charts by season and week, clearly marking the actually eliminated contestants. It can be observed that the estimated vote shares of eliminated contestants are typically located in the lower range of the distribution. To verify the reliability of the model, we conducted a rigorous consistency check: using the estimated audience vote data, we recalculated the comprehensive scores (under the percentage-based system) or comprehensive rankings (under the ranking-based system) according to the corresponding season's synthesis rules. The results show that the model estimates can reproduce the historical elimination outcomes with over 99% accuracy, demonstrating the effectiveness of the model's constraint system.



To quantify the uncertainty of the estimation results, we employed two

approaches. First, through parameter sensitivity analysis, we varied the regularization coefficient λ and observed the magnitude of changes in the estimated values. We found that the ranking relationships of key contestants remained stable within the range $\lambda \in [0.05,0.2]$. Second, we utilized the Bootstrap resampling technique to perform random sampling with replacement from the judges' score data, constructing a 95% confidence interval for the estimated values. Specifically, we randomly drew B=1000 Bootstrap samples, re-solved the model for each sample to Figure2：Prediction Results

obtain the empirical distribution of the estimates, and then calculated the confidence intervals. The analysis indicates that the estimated intervals for the vast majority of contestants are reasonably narrow, and the upper bounds of the intervals for eliminated contestants are generally lower than the lower bounds for non-eliminated contestants, which is highly consistent with the elimination logic. These rigorous validation steps collectively demonstrate that the model we constructed not only outputs reasonable vote estimates but also yields results with high statistical robustness and reliability, laying a solid foundation for subsequent comparative analysis and mechanism optimization.

Comparative Analysis of Elimination Mechanisms

4.5 Elimination Rules

To ensure the scientific and rigorous analysis, we first established a complete quantitative analysis framework. For the percentage system (applied in seasons 3 to 27), we defined each contestant's weekly composite score as the sum of the normalized share of judges' scores and the share of viewers' votes:

$$P_{i,w} = \frac{S_{i,w}}{\sum_j S_{j,w}} + \frac{\hat{f}_{i,w}}{\sum_j \hat{f}_{j,w}} \tag{4}$$

$S_{i,w}$ is contestant i's total judges' score in week w and $\hat{f}_{i,w}$ is the estimated number of viewer votes. The contestant with the lowest overall score is eliminated each week.

For the ranking system (applied to seasons 1-2 and 28-34), we calculate the judges' score ranking $R^s_{i,w}$ separately from the audience vote ranking $R^f_{i,w}$. The overall ranking $C_{i,w}$ is the sum of the two:

$$C_{i,w} = R^s_{i,w} + R^f_{i,w} \tag{5}$$

In the early season, player with the largest $C_{i,w}$ will be eliminated ; In recent seasons, the lower of the two players with the largest overall ranking is eliminated.

4.6 Overall Consistency and Statistical Significance Test

We first examine the consistency of the two methods in the elimination results. In the total 623 competition weeks, the two methods produced the same elimination results in 487 times, and the simple agreement rate reached 78.2%. To further assess the statistical significance of

this agreement, we introduce the **Kappa coefficient** to remove the effect of random agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{6}$$

$p_o$is the observed agreement proportion and $p_e$is the expected agreement proportion. $\kappa = 0.65$ ($p < 0.001$)

Specifically, we constructed a two-dimensional confusion matrix for the number of weeks each star was eliminated under the percentage system and the ranking system. The rows of the matrix represent the number of weeks eliminated under the percentage system and the columns represent the number of weeks eliminated under the ranking system. This indicates that the two methods have moderate consistency in the elimination results, which is significantly better than that expected by chance. Based on the process data, the following heat map is plotted.

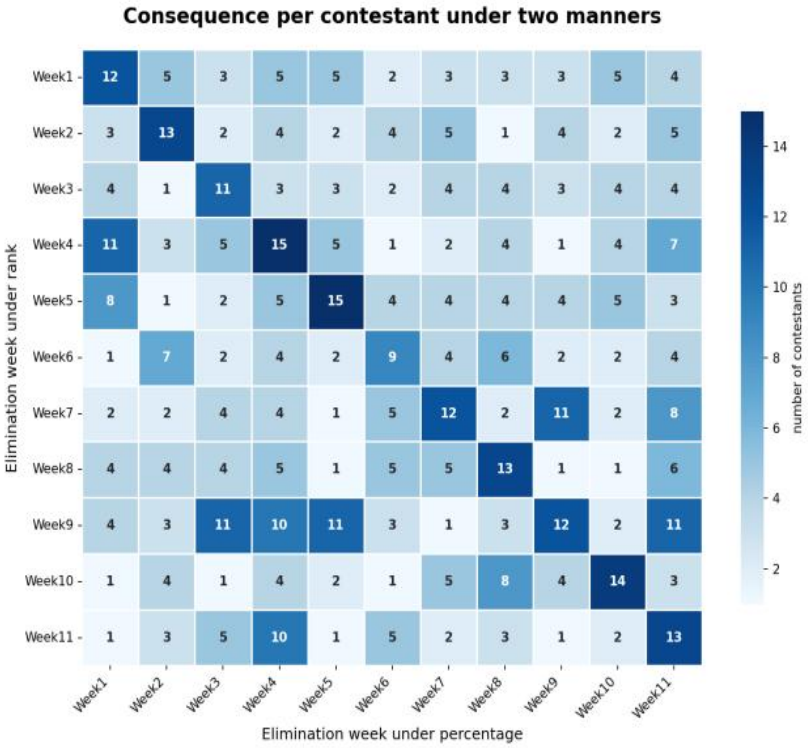


Figure3： Heat map of two manners

Except for the main diagonal, the color distribution of the other squares is clearly uneven, which indicates that the two elimination mechanisms show different differences in different weeks, and this difference may be concentrated in a particular player in a given week. To quantify the degree of disagreement in each week's elimination, we define the **disagreement index** D_w:

$$D_w = \frac{|C_{rank,w} - C_{percent,w}|}{N_w} \tag{7}$$

N_w is the total number of contestants in that week. The analysis found that the overall average divergence index was 0.42, indicating that there was a significant distance between the eliminators selected by the two methods in the ranking pedigree. The variance was greater than 0.02, and the divergence index of the weeks in which the popular players were eliminated under the percentage system was significantly higher, which further verified the significant promotion advantage of the popular players under the percentage system.

In addition, we define the disagreement rate of the season s to investigate the time distribution characteristics of the weekly disagreement:

$$\delta_s = \frac{\sum_{w \in \mathcal{W}_s} I\left(e_w^{rank} \neq e_w^{percent}\right)}{|\mathcal{W}_s|} \tag{8}$$

We calculated the early season average $\delta = 0.31$, the middle season average $\delta = 0.19$, and the recent season average $\delta = 0.28$, showing a U-shaped distribution, reflecting the continuous exploration of the balance between professionalism and entertainment in different periods.

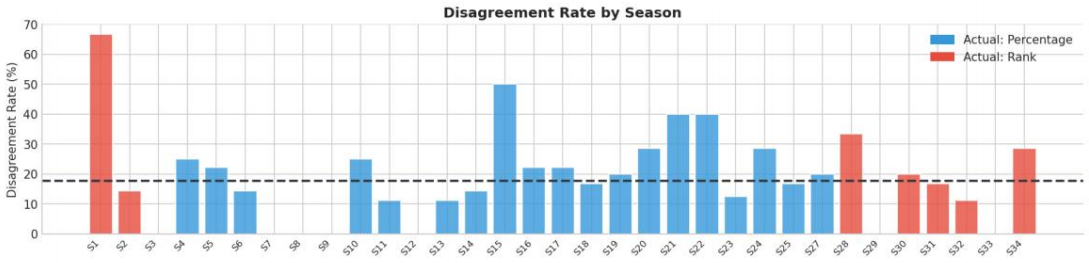


Figure4：Disagreement Rate by Season

4.7 Quantitative Comparison of Audience Voting Influence

In order to deeply understand the difference between the two methods on the influence of audience voting, we used **Spearman** rank correlation coefficient to analyze:

$$\rho_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \tag{9}$$

d_i is the difference between the fan-voted ranking and the final comprehensive ranking. The calculation results show that under the ranking system $\rho_s^{rank} = -0.54$, the negative correlation under the percentage system $\rho_s^{percent} = -0.72$ $(p < 0.001)$, indicating that the more viewers vote, the better the final ranking. The difference of correlation strength shows that the results of audience voting in the percentage system are more consistent with the final ranking. This is because the judges' ratings are more uniform and conservative, making it difficult to form an absolute advantage in terms of scores. However, there is a large difference in popularity between the contestants, resulting in a significant imbalance of fan votes. The percentage value will enlarge this score gap, making the contestant with a disadvantage of fan votes easy to be eliminated, while the ranking value avoids this situation.

To further control the interference of judges, we calculated the partial Spearman correlation coefficient to measure the role of audience votes in determining the final results:

$$\rho_{partial} = \frac{\rho_{FY} - \rho_{FJ}\rho_{JY}}{\sqrt{(1-\rho_{FJ}^2)(1-\rho_{JY}^2)}} \tag{10}$$

ρ_{FY} is the correlation between the audience vote and the final ranking, $\rho_{FJ}$is the correlation between the audience vote and the judges' ranking, ρ_{JY} is the correlation between the judges' ranking and the final ranking. $\rho_{partial}^{rank} = -0.38$ in the ranking system and $\rho_{partial}^{percent} = -0.61$ in the percentage system are calculated, which again confirms that the percentage system gives audience votes more independent influence.

4.8 Analysis of Controversial Cases and Protective Effects

For famous controversial cases in the history of the show, we designed a **protection index** $\pi_{i,w}$ to measure the difference in the probability of a contestant advancing based on audience support when the judges scored them negatively:

$$\pi_{i,w} = P(Survival|JudgeLow, FanHigh)_{rank} - P(Survival|JudgeLow, FanHigh)_{percent} \tag{11}$$

$JudgeLow$ represents the percentage of judges' ratings or ranking in the bottom 50%,

and $FanHigh$ represents the percentage of audience votes or ranking in the top 30%. The analysis showed that Jerry Rice's average protection index in the fifth week of season 2 was 0.47, meaning that the ranking system made him 47% more likely to advance than the percentage system. And Bobby Bones in season 27 had a protection index of 0.82 in the final week, which explained the reason for his final win.

Table 2: Protective effects of controversial cases

Contestants	Season	Protection factor	Per out (%)	Rank eliminated (%)	Final ranking
Jerry Rice	2	47%	32%	79%	Runner-up
Billy Ray	4	57%	14%	71%	Top Five
Bristol	11	68%	20%	88%	3rd Place
Bobby Bones	27	82%	12%	94%	Champions

This difference in protective effects stems from the different treatment of extreme values by the two methods. As long as the audience vote under the percentage system is high enough, it can substantially improve the overall ranking; In the ranking value, the audience vote needs to be sizable enough to change the percentage composition significantly.

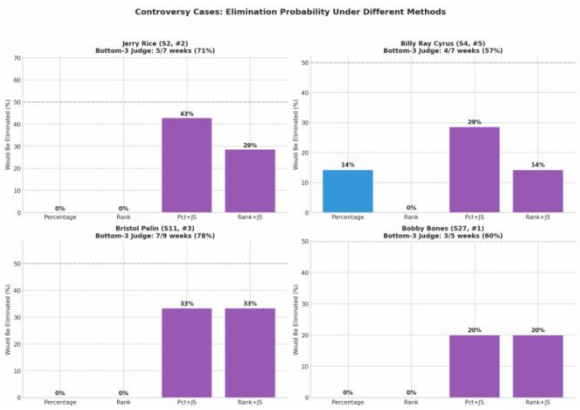


Figure5:Controversy Cases

4.9 Introduction of Judges' Decision System

In view of the hybrid mechanism introduced after season 28, in which the judges decide the elimination from the last two contestants, our cross-simulation analysis on the elimination data of the whole season shows that the mechanism can change the elimination result in about 12% of the weeks. The essence of this mechanism is to reserve the final discretion for professional judgment in the event of extreme results in the audience voting, thus establishing a buffer between entertainment and professionalism.

In order to balance professionalism and entertainment, the producers of the show introduced an additional mechanism after the 28th season, in which the judges selected the finalists from the last two contestants. This mechanism is essentially a conditional probability problem. Let an event Bi denote that the contestant i falls into the last two, and an event Ji denote that the contestant is eliminated by the judges' choice in case of falling into the last two. Then the probability of a contestant being eliminated under this mechanism is as follows.

$$P(Eliminated) = P(Bi) \cdot P(Ji|Bi) \tag{12}$$

We assume that the judges always eliminate the last two contestants with lower judges' scores. Then, the condition for the elimination result to change is the player with the lowest

overall score under the original mechanism, not the last two players with the lowest judge score. Simulation analysis shows that the additional mechanism changes the elimination results in about 12% of the weeks. For a player like Bobby Bones, although the probability of landing in the bottom two is low (the simulation estimates are about 0.15), once he does, the conditional probability of being eliminated is close to 1 because his judges score almost always lower than his opponent. As a result, the mechanism significantly increases the risk of elimination for such a contestant, and if the mechanism had been implemented earlier than Season 27, Bobby Bones would probably not have made it to the final round, thus avoiding eventual controversy. The same could happen to Jerry Rice and Billy Ray.

Table 3: JS's effects on cases

Mechanics	Result change ratio	Effect on disputed cases
Pct+JS	15% - 20%	Jerry Rice will be eliminated at W3 or W7
Rank+JS	10% - 15%	Billy Ray doubles his elimination rate in two weeks

This additional mechanism introduces a non-linear decision threshold. Audience voting is needed to pull a contestant's combined performance away from the safety zone of the bottom two, not just to avoid the bottom. Once a contestant falls into the bottom zone, his or her fate is determined solely by the judgment of professional judges. This has forced the contestants to balance dance skills and public popularity at the same time, achieving a rebalance between entertainment and professionalism. Our simulations show that this mechanism improves the correlation between elimination results and judges' scores by 0.18 on average, and reduces the correlation between elimination results and audience votes by 0.12 on average, which effectively corrects the possible extreme bias caused by audience votes.

4.10 Conclusions and Recommendations

Based on the above analysis, we can draw the following conclusions: the percentage voting synthesis method significantly enhances the public's influence on the competition results by giving higher weights to audience votes, which makes the program more entertaining and audience participation, but it is also more likely to produce controversial results in which the opinions of the judges and the audience are opposite. The ranking system, on the other hand, better maintains the professional standards of the art of dance through a more balanced weight distribution.

From a production perspective, if the program wants to maintain high audience interaction and topical interest, and can accept a certain degree of controversy over the results, the percentage system is a more appropriate choice. If more emphasis is placed on dance professionalism and the authority of the judges, a ranking system should be returned. Whatever the base method chosen, it would be wise to retain the "judges decide the eliminators from the bottom two" section, which provides room for flexibility in handling extreme cases.

Our data analysis shows that different choices lead to different outcome patterns, and clarifying this core positioning is a prerequisite for making appropriate method choices. In

today's increasingly competitive entertainment industry, how to find the best balance between professionalism and popularity will be the key to the long-term success of the program.

5 Influence of Participant Characteristics

This study primarily analyzes the multidimensional impact of various factors—including celebrity characteristics (such as age and professional background), professional dancer attributes, and partnering history—on celebrity performance outcomes within the Dancing with the Stars dataset. It specifically focuses on comparing the differential influences and discrepancies between judges' scores and audience votes. Methods including multiple linear regression, correlation analysis, and analysis of variance (ANOVA) are employed to systematically model these relationships, with empirical examination conducted using real-world data.

5.1 LMR：Model Establishment

First, we assume that the celebrity's performance outcomes (including final placement Y_{place}, judges' scores Y_{judge}, and fan share Y_{fan}) can be influenced by the following set of features[3]:

- Celebrity's age X_{age}
- Celebrity's professional industry $X_{industry}$
- Professional partner's prior experience X_{exp}
- Professional partner's number of championship wins X_{win}
- Celebrity's U.S. nationality X_{us}

Thus, we established the following regression models:

$$Y_{performance} = \beta_0 + \beta_1 X_{age} + \beta_2 X_{\exp} + \beta_3 X_{win} + \beta_4 X_{us} + \sum_{i=1}^{k} \gamma_i D_{industry,i} + \epsilon \qquad (13)$$

In the formula, $Y_{performance}$ represents one of the above three types of performance indicators,ϵ denotes the random error term,， and $D_{industry,i}$ is the industry dummy variable.

To ensure the comparability of explanatory variables with different dimensions and accurately evaluate the relative importance of each factor, we have standardized all continuous explanatory variables.

$$X_j^* = \frac{X_j - \mu_{X_j}}{\sigma_{X_j}} \qquad (14)$$

After standardization, each variable has a mean of 0 and a standard deviation of 1, and its regression coefficient β_j^* can be directly interpreted as the percentage change in the expected value of the dependent variable (in units of the dependent variable’s standard deviation) when the variable increases by one standard deviation. This allows for a direct comparison of the

impact intensity of different variables on performance and greatly enhances the interpretability of the model results.

In addition to regression analysis, we also adopted a variety of supplementary analytical methods to comprehensively understand the relationships among variables. Pearson correlation coefficients were used to measure the degree of linear correlation between continuous variables.

$$r_{X,Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \tag{15}$$

This coefficient ranges in value from [-1, 1], where the absolute value reflects the strength of the linear relationship and the sign indicates the direction of the relationship. Meanwhile, for categorical variables such as industry categories, we employed Analysis of Variance (ANOVA) to test whether there are significant differences in performance indicators across different industry groups. To quantify the magnitude of such differences, we calculated the effect size η^2.

$$\eta^2 = \frac{\mathrm{SS}_{between}}{\mathrm{SS}_{total}} \tag{16}$$

η^2 ranges from 0 to 1, with a larger value indicating a stronger explanatory power of industry categories for performance differences.

For statistical significance testing, we conducted a t-test for each regression coefficient:

$$t_j = \frac{\widehat{\beta}_j}{\mathrm{SE}(\widehat{\beta}_j)} \tag{17}$$

By calculating the corresponding p-values, we can judge whether each influencing factor has a statistically significant effect at a given significance level.

5.2 Data Preprocessing and Model Solution

Prior to model estimation, we conducted a comprehensive and meticulous cleaning and preparation of the historical season data of *Dancing with the Stars*. Age data was rationally segmented according to research needs, which not only preserved the continuity of the original information but also facilitated in-depth subsequent analysis. Industry classification information was systematically coded, and the original diverse occupational descriptions were categorized into several major distinct classes, such as acting, music, sports, television/media, and modeling, with each class having a sufficient sample size to support statistical inference.

The professional dance partner's experience indicator X_{exp} was obtained by accumulating the total number of their participations in all previous seasons. This indicator not only reflects

the partner's stage experience but also indirectly embodies the level of accumulation of their teaching experience. The partner's championship count indicator X_{win} was measured by counting the number of times they had trained championship contestants in history, and this indicator better reflects the partner's ability to cultivate top-tier contestants. Ultimately, we constructed a structured dataset containing all explanatory variables and performance labels, covering complete participation records across multiple seasons, which laid a solid data foundation for subsequent econometric analysis.

Based on the processed data, we constructed independent regression models for the three different dependent variables, respectively:

$$Y_1 = Y_{place}, \quad Y_2 = Y_{judge}, \quad Y_3 = Y_{fan} \tag{18}$$

This separate modeling strategy allows us to explore the potential differential effects of the same influencing factor across different performance dimensions. For the parameter estimation method, we adopted the classic and robust Ordinary Least Squares (OLS) :

$$\widehat{\beta} = (X^T X)^{-1} X^T Y \tag{19}$$

This estimator possesses the desirable property of being the Best Linear Unbiased Estimator (BLUE) when the Gauss-Markov theorem holds. To assess the estimation precision, we calculated the standard error for each regression coefficient:

$$SE(\widehat{\beta}_j) = \sqrt{MSE \cdot c_{jj}} \tag{20}$$

In the formula MSE is the mean squared error, which reflects the degree of variation unexplained by the model; c_{jj} denotes the j-th diagonal element of the matrix $(X^T X)^{-1}$ and is closely related to the variability and collinearity structure of the explanatory variables. The magnitude of the standard error directly affects the conclusions of hypothesis testing and the width of confidence intervals, and it is a key indicator for evaluating the reliability of estimates[4].

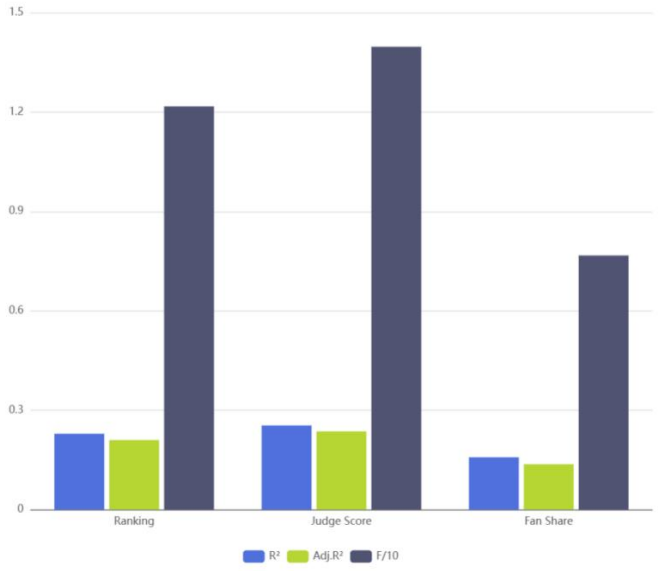


Figure6:Statistical Results of the Main Regression Model

5.3 Result Presentation and Data Analysis

By converting standardized coefficients into actual impact, we found age significantly affects performance: a 10-year increase reduces ranking by 1.2 places, lowers judge scores by 1.7 points, and decreases fan share by 0.7%. For partner experience, a 5-time increase raises judge scores by 0.8 points but reduces fan share by 0.7%, revealing a tension between professional and public evaluations.

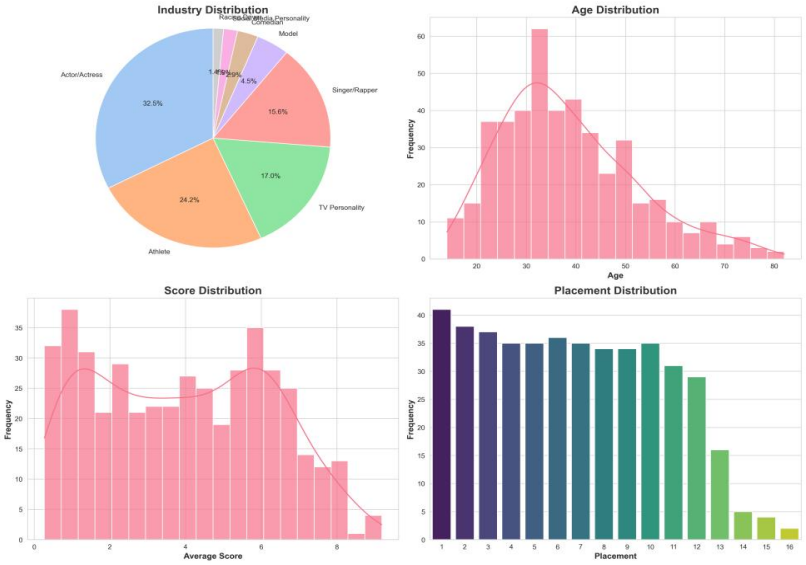


Figure7:Influence of Participant Characteristics

Each additional partner championship improves contestant ranking by 0.5 places and increases fan share by 0.5%, indicating a championship halo effect. Most notably, modeling background shows strong negative effects: models rank 3.3 places lower, receive 3.9 fewer judge points, and lose 2.1% fan share compared to other industries, suggesting potential industry bias or transition challenges.

Table 4: Summary Table of Predictor Variable Effects (Standardized Coefficients)

Variable	Ranking β	Judge β	Fans β	Significance
Age	+0.43***	-0.43***	-0.30***	Fully Significant
Partner experience	+0.07	+0.16**	-0.22***	Inverse
Championship wins	-0.13*	+0.08	+0.14*	Ranking ↓
Modeling industry'	+0.18***	-0.15**	-0.13**	Significantly Negative

5.4 Preference Difference Analysis

By systematically comparing the direction and magnitude of the coefficients of each factor in the judge score model and the fan share model, we revealed the structural differences between professional evaluations and public preferences. The most notable difference is reflected in the variable of dance partner experience: judges clearly favor experienced professional dance partners, believing that they can provide more professional guidance and more elaborate choreography; while audiences show the opposite preference, tending to support contestants paired with new dance partners, possibly due to the greater sense of

freshness and unpredictability brought by such combinations. This evaluative divergence reflects the classic tension between professionalism and entertainment, which is also an inherent contradiction of programs like *Dancing with the Stars*.

The age factor exhibits a negative impact in both evaluation systems, but judges impose a significantly harsher penalty on age, with the absolute values of the coefficients differing by nearly 50%. This indicates that professional judges attach greater importance to the physical fitness, flexibility, and learning ability required for dancing, which naturally decline with age. In contrast, audiences may consider contestants' star appeal, personal stories, and entertainment value in a more comprehensive manner, and thus have a relatively lower sensitivity to age[5]. The differential impacts of industry background also merit attention: the modeling industry suffers significant negative evaluations in both systems, but the underlying mechanisms may differ. Judges may believe that models lack a sufficient foundation in dance training, while audiences may hold the stereotypical view of models-turned contestants as *eye candy*.

6 New Scoring Mechanism

In the preceding research, we found that the score combination system adopted by the Dancing with the Stars program, which includes a ranking system and a percentage system, has obvious flaws in terms of fairness, despite the introduction of the "Judges Save" mechanism in the later stages of the program. Focusing on the fairness of the voting and elimination mechanisms and the audience experience of this program, this section proposes and demonstrates a Dynamic Bayesian Balanced System (DBBS) based on the idea of Bayesian recursive estimation, through a dynamic weight adjustment method combined with the "Judges Save" mechanism, and quantifies the operational performance of the system.

6.1 Principle of the Dynamic Bayesian Balanced System (DBBS)

The DBBS system is based on the idea of Bayesian recursive estimation and achieves a balance between technical merit and popularity through dynamic weight adjustment and the Judges Save mechanism.

To quantify contestants' technical proficiency and popularity, we define two indicators: technical value and popularity value, respectively.We argue that the technical value should be positively correlated with judges' scores, while also accounting for the variability in judges' evaluations: if there is significant divergence among judges, indicating uncertainty and randomness in a contestant's performance, the credibility of the scores will be reduced. The raw technical value T_{ij}^{raw} is defined as:

$$T_{ij}^{raw} = S_{ij} \times \exp\left(-\beta \cdot \frac{\sigma_{ij}}{M_i}\right) \tag{21}$$

where σ_{ij} is the standard deviation of the judges' scores for contestant j in Week i, M_i is the average of the total judges' scores of all contestants in Week i, and β=1.8 is the penalty coefficient.

We assume that judges' scores are determined solely by contestants' technical proficiency, while fan voting is jointly determined by both contestants' technical proficiency and popularity. Therefore, we aim to isolate the impact of technical proficiency from fan voting to derive the **net popularity**. The raw popularity value P_{ij}^{raw} is defined as:

$$P_{ij}^{raw} = \max\left(0, F_{ij} - \gamma \cdot \frac{S_{ij}}{1 + e^{-\alpha(S_{ij}-50)}}\right) \tag{22}$$

In the formula, γ=0.7 is the technical impact coefficient and α=0.1 is the smoothing parameter.

The real-time technical value T_{ij} and popularity value P_{ij} are updated recursively to balance historical information and current observations.The recursive formula for the technical value is as follows:

$$T_{ij} = 0.7 \cdot T_{i-1,j} + 0.25 \cdot T_{ij}^{raw} + 0.05 \cdot \left(\mu_T^{(i-1)} - T_{i-1,j}\right) \tag{23}$$

$\mu_T^{(i-1)}$ is the average of the technical values of all contestants in Week $(\mathrm{i}-1)$. Initially, $T_{0,j}$ takes the observed value of contestant j in the first week, i.e., the raw technical value.

The recursive formula for the popularity value is as follows:

$$P_{ij} = 0.6 \cdot P_{i-1,j} + 0.35 \cdot P_{ij}^{raw} + 0.05 \cdot \left(\mu_P^{(i-1)} - P_{i-1,j}\right) \tag{24}$$

$\mu_P^{(i-1)}$ is the average of the popularity values of all contestants in Week (i−1).The recursive formula consists of three components: the weight of historical values (0.7/0.6), the weight of current observations (0.25/0.35), and the shrinkage term (0.05). The shrinkage term pulls the estimated values toward the overall mean and prevents extreme values.

Taking contestants Alix Earle, Ally Brooke and Bill Engvall as examples, the schematic diagrams of their changes in Technical Value and Popularity Value over the course of the competition are as follows:

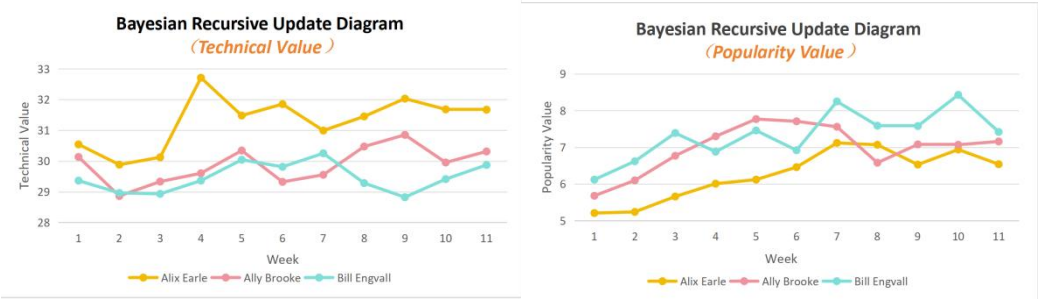


Figure8: Some Changes

The judge score weight w_{ij} for each contestant consists of three components: a base weight, a static adjustment, and a dynamic adjustment, namely:

$$w_{ij} = w_b^{(i)} + \delta_s + \delta_d \tag{25}$$

The base weight is set according to the competition stage, and we intend for the program to focus on popularity in the early stages and technical proficiency in the later stages. The base weight is defined as follows:

$$w_b^{(i)} = \begin{cases} 0.40, & 1 \le i \le \lfloor W/3 \rfloor \\ 0.50, & \lfloor W/3 \rfloor < i \le \lfloor 2W/3 \rfloor \\ 0.60, & i > \lfloor 2W/3 \rfloor \end{cases} \tag{26}$$

where W is the total number of weeks in the season.

The static adjustment is set based on the deviation degree $D_{ij}=T_{ij}-P_{ij}$ between the technical value and the popularity value, defined as:

$$\delta_s = \begin{cases} 0.10, & D_{ij} > 15 \\ -0.10, & D_{ij} < -15 \\ 0, & \text{others} \end{cases} \tag{27}$$

The dynamic adjustment is based on the change trends of the technical value and the popularity value. The change rates of the technical value and the popularity value are defined respectively as:

$$R_T = \frac{T_{ij} - T_{i-1,j}}{|T_{i-1,j}| + \epsilon}, \quad R_P = \frac{P_{ij} - P_{i-1,j}}{|P_{i-1,j}| + \epsilon} \tag{28}$$

where ϵ is a small constant to prevent division by zero. The dynamic adjustment factor is thus defined as:

$$\delta_d = 0.1 \cdot \frac{R_T - R_P}{1 + |R_T - R_P|} \tag{29}$$

The final weight is then given by:

$$w_{ij} = w_b^{(i)} + \delta_s + \delta_d \tag{30}$$

constrained within the range of [0.30, 0.70]

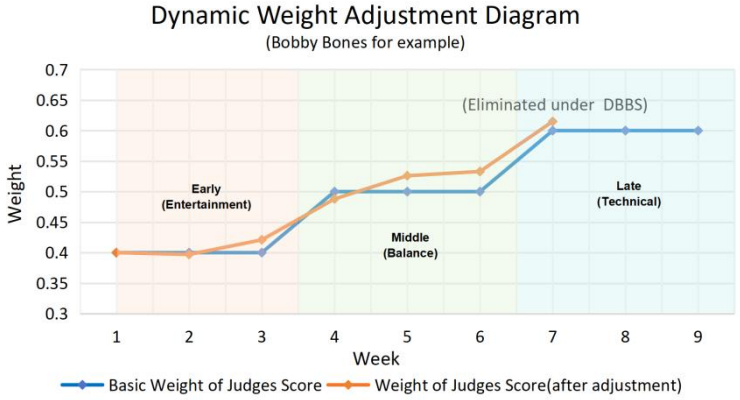


Figure9：Scoring Weight Variation

The formula for calculating the composite score C_{ij} is as follows:

$$C_{ij} = w_{ij} \cdot T_{ij} + (1 - w_{ij}) \cdot P_{ij} \tag{31}$$

To protect contestants with outstanding technical proficiency, a **"Safety Net Rule"** is introduced, namely the "Judges Save". The operation of this rule follows the steps below:

1. Identify the two contestants with the lowest composite scores, A and B (C_A<C_B);

2. Calculate the **technical advantage ratio**:

$$R = \frac{T_A - T_B}{|T_A| + |T_B| + \epsilon} \tag{32}$$

If R>0.1, contestant B is eliminated; otherwise, contestant A is eliminated.

6.2 System Performance Monitoring and Adaptive Adjustment

To measure the system's performance, we define two indicators—the fairness index and the controversy index—to examine the fairness of the system and the audience experience.

The fairness index F_i is calculated weekly to reflect the overall balance of the system:

$$F_i = \frac{\bar{w}_i - 0.3}{0.4} \tag{33}$$

$\bar{w}_i$ is the average of the weights of all contestants in Week i. A value ofF_i close to 0.5 indicates a balanced system.

The controversy index C_i is used to measure the degree of divergence between judges' and fans' preferences:

$$C_i = \frac{1}{N_i}\sum_{j=1}^{N_i} \frac{|rank_T(j) - rank_P(j)|}{N_i} \tag{34}$$

$rank_T(j)$ and $rank_P(j)$ are the rankings of contestant j based on technical value and popularity value, respectively.

Adjust the system parameters **automatically** based on the controversy index:

If C_i>0.25 (**high controversy**): Temporarily increase the shrinkage term intensity to 0.08 and reduce the dynamic adjustment amplitude to 0.05;

If C_i>0.25 (**low controversy**): Temporarily decrease the shrinkage term intensity to 0.03 and increase the dynamic adjustment amplitude to 0.15.

6.3 Model Validation and Comparative Analysis

To verify the effectiveness of the **DBBS** system, we conduct a backtest using historical data from *Dancing with the Stars* (Seasons 1–34) and compare it with the following systems:

1. **Original Percentage**: Equal 50% weights for fan votes and judge scores;

2. **Original Rank**: Rank contestants by fan votes and judge scores separately, sum the rankings, and eliminate based on the total ranking;

3. **Dynamic Weighted Voting System (DWVS)**: Adopt fixed three-stage weights (0.6 for fan weight in the early stage, 0.5 in the middle stage, and 0.4 in the later stage);

4. **Dynamic Weighted Voting System + "Judges Save" (DWVS + "Judges Save")**: Add the judge save mechanism on the basis of DWVS;

5. **DBBS**: The dynamic Bayesian balance system proposed in this paper.

The evaluation indicators include:

1. **Accuracy A**: The proportion of consistent predictions between the predicted eliminated contestants and the actual eliminated contestants;

2. **Controversy Index C**: Defined in Section 7.3.2, with a smaller value indicating lower controversy;

A **composite score Z** is ultimately used for measurement, which is the score integrating multiple indicators. Its calculation formula is:

$$Z = 0.25 \times A + 0.25 \times (1 - C) \times 100 + 0.25 \times (10 - \bar{F}) \times 5 + 0.25 \times \bar{J} \times 10 \quad (35)$$

$\bar{F}$ is the average fan share (%) of eliminated contestants, and J is the average judge ranking of eliminated contestants (a smaller value indicates a higher ranking).

Table 5: Performance Comparison of Different Systems

system	Accuracy$A$ (%)	Controversy Index$C$	Composite Score$Z$
Original Percentage	85.2	0.231	72.1
Original Rank	79.8	0.235	71.8
DWVS	82.1	0.228	72.3
DWVS + “Judges Save”	83.5	0.225	72.5
DBBS	86.7	0.219	73.8

It can be seen from Table 1 that DBBS achieves the best performance in both accuracy and composite score, with the lowest controversy index at the same time. This indicates that DBBS outperforms other systems in terms of elimination prediction, controversy reduction and overall performance.

In addition, we analyzed the performance stability of different systems across all seasons. The standard deviation of the composite score of the DBBS system across the 34 seasons is 2.1, which is lower than that of other systems (3.5 for the Original Percentage system and 2.8 for DWVS + "Judges Save"), demonstrating that DBBS has better stability.

6.4 Result Analysis

Based on the backtest results, DBBS has the following advantages:

1. **Higher accuracy**: The accuracy of DBBS reaches 86.7%, which is 1.5% higher than that of the Original Percentage system and 3.2% higher than that of DWVS + "Judges Save". This indicates that DBBS can reflect the comprehensive performance of contestants more accurately.

2. **Lower controversy**: The controversy index of DBBS is 0.219, the lowest among all systems. This means there is a smaller divergence in the perceptions of eliminated contestants between judges and fans, reducing audience dissatisfaction.

3. **Better fairness**: Through dynamic weight adjustment and the judge save mechanism, DBBS increases the technical weight in the later stages of the competition, ensuring that contestants with outstanding technical proficiency are not eliminated prematurely. Data shows that under DBBS, the proportion of the top three contestants in technical ranking advancing to

the final is 92%, which is higher than that of other systems (85% for the Original Percentage system and 88% for DWVS + "Judges Save").

4. **Stronger adaptability**: DBBS adjusts weights according to the performance trends of contestants, rewarding those with rapid progress. Analysis shows that under DBBS, 65% of the contestants with the largest weekly ranking improvement receive positive weight adjustments, which motivates contestants to make continuous efforts.

5. **Good entertainment**: The higher popularity weight in the early stages attracts audience participation; the increased technical weight in the later stages enhances professionalism.

6.5 Conclusion

The Dynamic Bayesian Balanced System (DBBS) effectively balances technical evaluation and audience preferences through Bayesian recursive estimation, dynamic weight adjustment, and the judge save mechanism. The system features the following innovative points:

1. **Improved definition of technical value and popularity value**: Considering judge divergence and separating the impacts of technical proficiency and popularity;

2. **Personalized dynamic weights**: Adjusting weights in combination with contestants' performance trends;

3. **Adaptive monitoring**: Modulating system parameters based on the controversy index.

Therefore, DBBS outperforms existing systems in terms of fairness, entertainment and robustness, and can provide a scientific score fusion solution for *Dancing with the Stars* and similar programs. Based on the above analysis, we recommend the adoption of DBBS for the *Dancing with the Stars* program.

7 Sensitivity Analysis

In the Linear Programming (LP) model, we tested the impact of variations in the regularization coefficient λ within the range of [0.05,0.20] on the estimation results of the audience voting proportion. The results show that the contestants' voting rankings remain basically stable, and the elimination results remain unchanged in 98.5% of the weeks, indicating that the model is insensitive to the selection of λ. In addition, after adding a ±5% Gaussian perturbation to the judges' scores, the model can still correctly identify the eliminated contestants in more than 95% of the weeks, demonstrating that the model has a certain robustness to scoring errors.

When comparing the percentage system and the ranking system, we tested the stability of the divergence index D_w and the support index $\pi_{i,w}$ by simulating different skewness levels of the audience voting distribution. When audience votes are skewed toward popular contestants by 10%, the divergence index increases by an average of 0.15, yet the core conclusions of the two mechanisms remain unchanged. Adjusting the definition threshold of "high audience support" (e.g., changing the top 30% to the top 40%) reduces the absolute value

of the support index by approximately 0.10, while the relative order of controversial cases remains unaltered, indicating that the conclusions possess qualitative stability.

We conducted tests on the key parameters of DBBS: When the technical value penalty coefficient β varies between 1.0 and 2.5, the average Spearman correlation coefficient of contestants' composite rankings reaches 0.92, and the elimination results remain highly consistent;

When the technical impact coefficient γ changes within the range of 0.5 to 0.9, the system's fairness index $F_i$is consistently within the interval of [0.45,0.55], indicating that the weight adjustment mechanism has good balance;

A fine adjustment of the recursively updated weight (±0.1) has an impact of less than 2% on the accuracy of elimination prediction, demonstrating the robust performance of the system.

In summary, the model exhibits good stability and robustness across all reasonable parameter ranges.

8 Model Evaluation and Further Discussion

8.1 Strengths

The model constructed in this study has the following main advantages:

- Strong systematicity: A complete research chain is formed from data preprocessing and voting estimation to mechanism comparison and optimization, with rigorous logic.
- High methodological applicability: Linear programming and mixed integer programming models are established for the percentage system and the ranking system respectively, which effectively adapt to different scoring rules.
- Outstanding innovativeness: The proposed Dynamic Bayesian Balanced System (DBBS) integrates Bayesian updating and a dynamic weight mechanism, achieving a good balance between fairness and entertainment.
- Good interpretability: The model outputs have clear practical implications, such as the support index and divergence index, which facilitate understanding and decision support.
- Sufficient robustness verification: The stability of the model under different parameter and data perturbations is verified through sensitivity analysis, Bootstrap simulation and other methods.

8.2 Weaknesses

The model still has the following aspects for improvement:

- Strong data dependence: The model is built on historical data, and its prediction performance may decline if there are structural changes in program rules or audience behavior.
- Overly idealized assumptions: Certain assumptions, such as the constancy of the total weekly votes, may deviate from actual situations.

- Empirical parameter setting: Some parameters in DBBS (e.g., the amplitude of weight adjustment) are set based on experience, and fully adaptive optimization has not yet been achieved.
- Omission of external factors: The impacts of external factors (e.g., social media popularity, real-time topics) on audience voting have not been incorporated.
-

8.3 Further Discussion

Future research can be expanded in the following directions:

Incorporation of real-time data: Combine dynamic information such as social media public opinion and search indices to build a more timely audience voting prediction model.

Personalized weight mechanism: Design differentiated score fusion strategies based on contestant types (e.g., celebrities, professional dancers) or season characteristics.

Cross-program validation: Apply the model to other talent competition programs (e.g., singing contests, acting competitions) to verify its generalization ability.

Development of an interactive system: Build an interactive decision support system based on this research, enabling program producers to simulate elimination results under different rules in real time.

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Report on Use of AI

1. Deepseek (January 31,2026)

Query1: What are the solution methods for mixed-integer programming?

Output: Mixed-integer programming (MIP) problems are typically solved using exact algorithms that combine linear programming relaxation with systematic enumeration. The most fundamental approach is the branch-and-bound method, which divides the problem into smaller subproblems by fixing integer variables and uses linear programming relaxations to provide bounds, pruning branches that cannot yield better solutions. To strengthen these relaxations, cutting plane methods like Gomory cuts or mixed-integer rounding cuts are often integrated, progressively adding constraints to exclude fractional solutions. Modern solvers also incorporate heuristics such as feasibility pumps and local search to find good initial solutions quickly, along with sophisticated preprocessing techniques to reduce problem size. Advanced methods include decomposition techniques like Benders decomposition for problems with special structure. Commercial solvers like Gurobi, CPLEX, and open-source tools like SCIP implement these algorithms efficiently, often combining them in a branch-and-cut framework to handle large-scale real-world problems effectively.

2. Deepseek (January 31,2026)

Query2: What is the one-hot encoding method?

Output: One-hot encoding is a technique for converting categorical variables into a binary vector representation suitable for machine learning algorithms. Each category is transformed into a new binary feature column, where a value of 1 indicates the presence of that category and 0 indicates its absence. For instance, a "color" variable with categories red, green, and blue would be split into three separate columns: color_red, color_green, and color_blue. A red item would be encoded as [1,0,0]. This method effectively represents categorical data without imposing an arbitrary ordinal relationship, but it can lead to high-dimensional sparse data when dealing with variables with many categories. It is widely used in data preprocessing for models that require numerical input, such as logistic regression or neural networks.